CAREERTRONIC
GLOBAL SERVICES (P) LTD

INDUSTRIAL AUTOMATION CERTIFICATION PROGRAM

www.careertronic.com

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CAREERTRONIC
GLOBAL SERVICES (P) LTD

ABOUT COURSE

COURSE OVERVIEW

Industrial Automation has become an essential part of every industry. Our industrial automation course helps students gain in-depth hands-on experience in industrial automation.
Automation helps industry to increase the quality, accuracy and precision of industrial processes. It helps increase productivity. The need for automation will always increase. The automation industry creates many job vacancies. It covers the entire manufacturing and processing sector.

WE TRAIN ON

SIEMENS

ABB

Allen-Bradley by ROCKWELL AUTOMATION

Schneider Electric

DELTA

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SIEMENS
Schneider Electric
ABB
Honeywell
Allen-Bradley by ROCKWELL AUTOMATION
YOKOGAWA
MITSUBISHI ELECTRIC
OMRON
DELTA

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CAREERTRONIC
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COURSE TOPICS

COURSE CONTENT

Our industrial automation training starts from the basic level and ends up at an advanced level. We deal with multi-brand of Equipment and controllers. All the brands' industries currently using.

- Introduction to Industrial Automation
- Instrumentation & Sensor Basics
- PLC – Programmable Logic Controller
- Motors & Variable Speed Drives
- HMI & SCADA
- PLC & HMI Programming
- Factory Automation Applications
- Process Automation Applications
- Process Controllers – Single & Multi-loop PID Controllers
- DCS Systems
- Industrial Data Communication and Networking
- Industrial IoT & Infrastructure Automation Applications

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CAREERTRONIC
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COURSE TOPICS

TOPIC TITLE	SUB-TOPICS
Introduction to Industrial Automation	• What is Automation? • What are the different components used in automation? • What are the different control systems used in automation?
Instrumentation & Sensor Basics	• Measurement basics • Widely used sensors in FA & PA • Process instrumentation fundamentals – • Temperature, Pressure, Flow, Level • Machine monitoring sensors – Load cell, Speed transducers, Proximity sensors • Control elements like Solenoid valves, Control Valves
PLC – Programmable Logic Controller	• Introduction and evolution of Programmable Logic Controller (PLC) • Architecture of a PLC • Elements of PLC • PLC Family – Nano, Micro, Small & Large & • evaluation of PLC power • PLC Configuration – Simplex & Redundant • Programming console • Earthing and Protection circuits • Limitations of PLC

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COURSE TOPICS

TOPIC TITLE	SUB-TOPICS
Motors & Variable Speed Drives	• Understanding of Motor and terminology • Primary purpose of Motor Control • Motor Starting Solutions • Introduction to AC Drive • Types of AC Drive • Introduction to DC Drives • Electrical switchgears & Panel design
HMI & SCADA	• Concept of HMI & SCADA • Hardware HMI types & selection • PC based HMI • Industrial Data Communication • Features of SCADA • Configuring Graphics, Alarms, Trends & Reports
PLC & HMI Programming	• Concept of IEC 61131-3 & PLC programming • Programming devices • Understanding and using Ladder, Function Block, Structured Text, Sequence Flow Chart • Diagnostics & Troubleshooting • HMI Programming • SCADA Programming

COURSE TOPICS

TOPIC TITLE	SUB-TOPICS
Factory Automation Applications	• Plastic Injection moulding & extrusion machine • Pharmaceutical machine • Textile machine • Packaging machine
Process Automation Applications	• Steel plant applications • Chemical plant applications • Power plant applications • O&G applications • Sugar plant applications • Pulp & Paper applications
Process Controllers – Single & Multi-loop PID Controllers	• Understanding PID – Mother of all controls • Overview of various control strategies – On-Off, Manual Tuning, Fuzzy Logic, Auto-Tune, Self Tune... • Overview of Control Algorithm - Feedback, Feed-forward, Cascade, Ratio... • Capabilities of Multi-Loop Controllers, its • features and programming

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COURSE TOPICS

TOPIC TITLE	SUB-TOPICS
DCS Systems	• Introduction and evolution of DCS System • DCS System hardware (and firmware) • DCS system software and protocols • Basic DCS Controller Configuration • Introduction to communication for DCS & OIS • Local Area Network Systems • Integration of complete fieldbus system • Programming of DCS systems • Alarm System Management • DCS system reporting
Industrial Data Communication & Networking	• What is industrial Data Communication? • What is a Network? • Examples & Types of Network • Network Topologies • Selection criteria of suitable network for given application • Interoperability between different networks • Media, Switch and Protocol Converter usage • Key Applications of popular network types

COURSE TOPICS

TOPIC TITLE	SUB-TOPICS
Industrial IoT & Infrastructure Automation Applications	• What is Industrial IoT? • How IoT works & its enterprise view • Industrial IoT Framework • IIoT Hardware – Gateways, IO Modules, Remote Access, Cloud-Box • 3C – Connect, Collect, Consume • Application Notes • Case studies of installed applications
Projects 1 & 2	• Design & simulation of control systems • Design & implementation of control scheme remote monitoring • Project Execution planning-theory • Projects Documentation • Basic safety in Sites • Professional Grooming sessions • Basic lab • Interview skill training. • Industrial Live Project

CAREERTRONIC
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100% JOB PLACEMENT ASSISTANCE

We have trained & certified 300+ students.

Recently placed students at highly growing companies like Adani Industries, Reliance Industries, Rockwell Automation, Siemens, Grandiose Automation, Phillips, Tata Projects Ltd, SSM Infotech, NMDC Jagdalpur, Vedanta Industries, MSP Steel & many more.

Fuji Electric
Innovating Energy Technology

adani
Power

TATA PROJECTS
Simplify.Create

MITSUBISHI ELECTRIC

SIEMENS

SSM

GRANDIOSE
LEADING SOLUTION PARTNER

Rockwell Automation

MSP
TMT BARS | STRUCTURALS | PIPES

ABB

Schneider Electric

PHILIPS
Engineering Solutions

Honeywell

JSW

vedanta
transforming for good

Reliance
Industries Limited

AB
QUALITY

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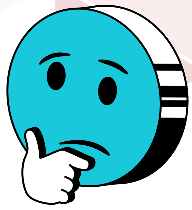
WHY JOIN US?



- Hands-on Training on 6 Leading Brands **Siemens, ABB, Allen Bradley, Schneider Electric, FUJI & Delta** etc
- Training by **Industry Experts**
- FREE Online Course **Access for 1 year** with class recording & notes
- Globally Recognised **Certification**
- **Job Placement Assistance**

PROGRAM DETAILS

Program Duration
6 Months Training +
6 Months (Additional Bonus Access)

Eligibility Criteria
Any graduate/post-graduate/ITI or diploma in EE, ECE, ETC or mechanical/mechatronics or instrumentation

Program Fee
Total Fees: ₹ 60,000 + GST (OFFLINE)
₹ 42,000 + GST (ONLINE)

GLOBALLY VALID CERTIFICATION

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FAQs

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1. Who can take up this course?

The course is specifically designed for:

- Engineering students doing bachelor and master degree who wish to expand their knowledge in Automation
- Industrial persons and faculty members who would like to develop capabilities in Automation
- Individuals seeking career in domains in Industrial automation applications
- Graduates who seek job in electrical, instrumentation, automation domain. The basic requirements to opt this course are as follows: One should be familiar with the basics of electrical & electronics. The knowledge of basic power electronic can be beneficial.

2. TOP RECRUITERS FOR THIS DOMAIN SKILL SETS NEEDED FOR STUDENTS TO GET PLACED WITH THESE RECRUITERS

- ABB/Hitachi India Design and maintenance of PLC & its operation, understanding of inverter topologies, development of control schemes for various applications
- Alstom Design and maintenance of PLC & its operation, understanding of inverter topologies, development of control schemes for various applications
- Siemens, CG drives, General electric Design and maintenance of PLC & its operation, understanding of inverter topologies, development of control schemes for various applications

3. What is included in your course?

This course is designed to include all requirements for a power electronic / Automation engineer or those required for research level jobs. It starts with introduction to basic in Instrumentation, inverters and then explains the applications, programming and diagnostics of PLC, HMI, SCADA & DCS systems. which will be directly helpful to opt for a job in this field or also to do a reseach or post graduation in the relavant field.

4. What will the student gain from your course?

With the evolution of automation technologies, the importance of Instrumentation, Control and Automation usage is increased significantly. Therefore, it is essential for an electrical / instrumentation engineer to understand this field thoroughly. In this course, students will get detailed theoretical knowledge and design insights with their control schemes. With this knowledge, students will be able to design, simulate and analyze the machine or process better. Further, with the knowledge of advanced control schemes covered in this course any new configuration for advanced state-of-the-art applications can be designed efficiently.

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FAQs

5. What software skills are you teaching and how well are these tools used in the industry?

This course cover relevant software tools viz MATLAB / Ltspice, relevant programming software of PLC/HMI & SCADA. MATLAB is predominantly used in the industry for analysis of power electronic circuits. While, Ltspice is an open-source software tool, which is exactly similar to Pspice/Simetrix used in most of the industries. Thus, students will be trained in multiple software tools. They will also be given mini projects involving designing and analysis of the circuits using these software tools in order to develop proficiency in these software tools. Students can use this knowledge in any industry.

6. What is the real world application for the tools and techniques will you teach in this course?

The course content is designed to cover real world applications of PLCs in simple machine application to complex process automation. In this course, students will be learning various applications used every day in real life. The analytical methods and software tools taught in this course can be directly applied to the architecture in these applications.

7. How is your course going to help a student's path to MS or PhD?
This course is designed to not only cover the applications of the PLC, SCADA and DCS System but it also emphasize on the basic concepts in the field of instrumentation and inverter systems. Further, the techniques to explore new configurations are also discussed, which will directly help for reseach field. Typically for MS and PhD these skill sets are required and thus, this course will definately help student to gain tremendous focus on the reseach path. As well as it will help to crack any entrance exam questions related to these field.

8. How is this course going to help a student get a job?

As mentioned earlier as well, this course is designed to not only cover the basic concepts but also applications in the industrial systems. Further, from basic switching mode converter to details on the modulation scheme principle, all basics are covered in detail. Additionally, the techniques to control the industry scale products are also discussed. The challenges and projects given in this course, which students will be solving are indegineously designed to train them in handling any industrial problem. Therefore, the skill sets obtained by the student as a part of this course will help him to not only crack the entrance or technical interview for such jobs but also to lead any industrial challenge as a part of his job profile related to this field.

FAQs

9. What are the job opportunities in this field?

Today, Automation components are prominently used in majorly all industrial system since the industrial revolution in Industry 4.0 has taken by storm. The major players in this area are ABB, Siemens, Fuji, Rockwell, Emerson, Mitsubishi, Alstom, Hitachi etc. These companies supply and use various products like PLC, HMI, DCS, SCADA, HMI, IIoT, Field Instrumentation, Analyzers etc. The techniques taught in this course will be directly applied to design and analyse these systems and thus in above mentioned industries.

10. How will I access the softwares required for programming?

We provide free use version softwares for practice to the students, which they can download in their laptops & practice at their own convenience.